CHAPTER

16

Cast Metal Restorations

Many clinical situations warrant the clinician to opt for cast metal restoration. These provide long-lasting restorations and maintain the proximal contact for a considerable period of time. The contacts and contours can also be modified with the help of these restorations.

NOBLE VS BASE METALS

The metal restorations are generally fabricated from either noble metals or base metals. 'Noble' refers to metals with marked resistance to oxidation and chemical reaction. Sometimes, the term 'precious' is also used. Precious usually refers to economic value. 'Base metal' refers to the metal elements that are chemically reactive to their environment. Recently, titanium has also been used for casting.

Noble metals can be gold based (gold, silver, copper along with platinum, zinc, iridium) or palladium based (palladium, silver, gold along with zinc, indium, iridium). The base metals are mostly cobalt based (cobalt, chromium, tungsten along with copper, nickle and gallium) or nickel based (nickle, chromium, iron along with boron, copper and manganese).

Taggart (1907) was credited for describing casting procedures in cast restorations.

The properties of individual constituents of cast metals are summarized in Table 16.1.

Amongst cast metals, gold alloy is most common. The gold alloy is divided into four types.

- Type I, Type II } Gold percentage between 75 and 80% to preserve the noble metal properties
- Type III, Type IV } Gold percentage between 65 and 75%

Rest is silver, copper, platinum, zinc, etc. in small percentage.

Types I and II are commonly used where stress-bearing area is less; whereas, Types III and IV are used in more stress-bearing areas.

The base metals are also used for cast restorations; however, disadvantages of base metal alloy are microleakage, difficulty in obtaining finer details and finishing and polishing.

Table 16.1: Properties of individual constituents

Gold	Adds high corrosion resistance, ductility and good castability
Silver	Reduces density, whitens the alloy counteracting redness of copper
Copper	Strengthens gold-based alloys
Platinum/palladium	Increases casting temperature, strength and corrosion resistance. Palladium lowers the cost and increases rigidity
Zinc	Increases castability
Iridium	Acts as grain refiner
Indium	Serves as bonding agent in metal ceramics
Gallium	Contributes the homogenous structure, improves castability
Cobalt	Provides strength, hardness and corrosion resistance
Chromium	Provides hardness and resilience and increases corrosion resistance to some extent
Nickel	Increases ductility and lowers melting temperature
Manganese	Acts as deoxidizer
Boron	Decreases alloy melting temperature

Titanium alloys are also being used for cast restorations. These are biocompatible; however, disadvantages are microleakage and difficulty in finishing and polishing.

Indications of Cast Restoration

1. *Extensive tooth involvement:* Cast restorations are efficient in replacing lost tooth structure and also for supporting remaining tooth structure (Fig. 16.1 A, B).
   Fractured teeth/marginal ridges are also indications for cast restorations.
2. *Adjunct to periodontal therapy:* Cast restorations can better restore and maintain dimensions of the contact, contour, marginal ridges and embrasure areas, which are vital for health of the periodontium (Fig. 16.2).
3. *Modifying contacts and contours* (superior control over contacts and contours): Cast restoration can reshape the deviated contact, contour and embrasures.
4. *Correction of occlusion:* If any change for occlusal table is planned, cast restorations are ideal.
5. *Endodontically treated teeth:* Most of the endodontically treated teeth have extensive tooth loss; need cast restoration.
6. *Retainers* for fixed prostheses
7. *Subgingival lesions:* In cavities with subgingival margins (Fig. 16.3) and where the prolonged exclusion of moisture is difficult, the preferred restorative material is the cast gold.
8. *Fracture lines:* Fracture lines in enamel, especially in teeth having extensive restorations are ideal for cast restorations (full crowns, onlays).





Fig. 16.1: Extensive tooth loss in mandibular molar



Fig. 16.2: Deviated contact in mandibular first molar

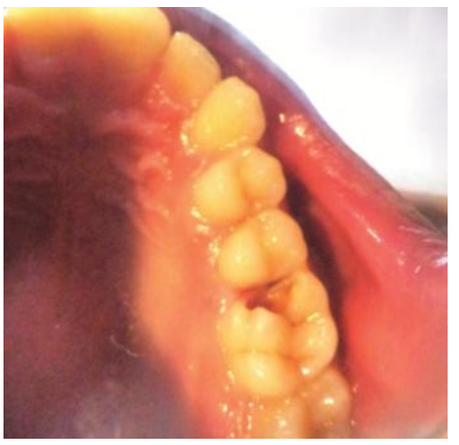


Fig. 16.3: Subgingival lesion on maxillary first molar

9. *Oral hygiene:* The tooth–cement–cast interphase is the weak link in cast restorations; patients performing adequate hygiene measures are indicated for cast restorations.

Contraindications of Cast Restoration

1. *Developing and deciduous teeth:* Young dentition with large pulp chambers are contraindicated for cast restorations.
2. *High plaque/caries indices:* Patients with rampant caries and poor oral hygiene should not be given cast restoration.
3. *Occlusal disharmony:* Cast restorations should not be used in patients with severe occlusal interference or other defects in the stomatognathic system.
4. *Dissimilar metals:* Gold-based castings are avoided in patients already having silver restorations.
5. Where esthetics is prime consideration, because metallic restorations display metal color.

Advantages of Cast Restoration

- Compressive strength, tensile strength and shear strength of alloys used for cast restorations are far greater than those of amalgam.
- Cast techniques reproduce precise form and minute detail; maintain the detail even under functional stresses.
- Gold cast restorations are not affected by tarnish and corrosion processes in the oral environment.
- Cast restorations provide fewer voids, less internal stresses, and maximum bonding between the component phases.
- Finishing and polishing is better.
- No volume change after placement.
- Gold castings have a coefficient of thermal expansion (12×10^{-6}/°C); approximately similar to that of tooth structure.
- Better control of contacts and contours.

Disadvantages of Cast Restoration

- Being a cemented restoration, several interphases will be created at the tooth–cement–casting junction. Leakage under a cast restoration is alarming; more pronounced gingivally.
- Cast restorations necessitate extensive tooth involvement, which may create hazards for the vital dental tissues.
- The cathode nature of cast gold alloys towards amalgam may lead to galvanic deterioration of amalgam, if these two restorations are placed adjacent to or opposite to each other.
- Cast alloys usually have a high abrasive resistance, more than that of tooth enamel (may abrade the opposing tooth and create occlusal imbalance).
- Technique sensitive.
- Comparatively expensive.

CAVITY DESIGN FOR CAST RESTORATIONS

The factors that influence design of the cavity are:
- Length of the clinical crown
- Anatomic characteristics of the occlusal, proximal, buccal and lingual surfaces
- Position of the tooth in the arch
- Occlusal and proximal relationships
- Esthetic issues
- Relationship and condition of the soft tissues
- Extent and location of the carious lesions

The conventional Black's principles are followed for preparing cast restorations.

1. Outline Form

a. *External Outline Form*

The external outline form for the gold inlay is, by and large, similar as that for amalgam cavity. The placement of bevels and taper make the outline form slightly wider for cast restorations. The beveled cavosurface margin is placed in sound tooth tissue (Fig. 16.4A to G).

b. *Internal Outline Form*

The amount of taper varies with depth of the preparation. The taper should not be visible to the eye. In shallow preparations, parallelism enhances the resistance and retention form of the preparation. Deep cavities require taper to facilitate seating of the restoration.

The pulpal floor is positioned 1.75 to 2.0 mm into dentin below the central groove (0.5 mm into DEJ).

The cervical floor should be in sound tooth tissue.

The cervical floor is positioned after caries removal (Fig. 16.5A). The bevel is placed to establish the cervical finishing line (Fig. 16.5B).

Line angles in both the occlusal and proximal portions of the preparation should be well defined. The axiopulpal line angle is slightly rounded (Fig. 16.6A). The flare of the proximal walls should form axioproximal angles of 100° to 110°.

2. Resistance and Retention Forms

Lateral/tangential forces may cause displacement of the restoration. Frictional retention can be achieved by length of dentin and enamel walls grasping the restoration (intracoronal retention). The cement bond alone will not provide retention for the casting. Cement provides only a moderate mechanical lock between the minute irregularities of the cavity walls and the casting surface.

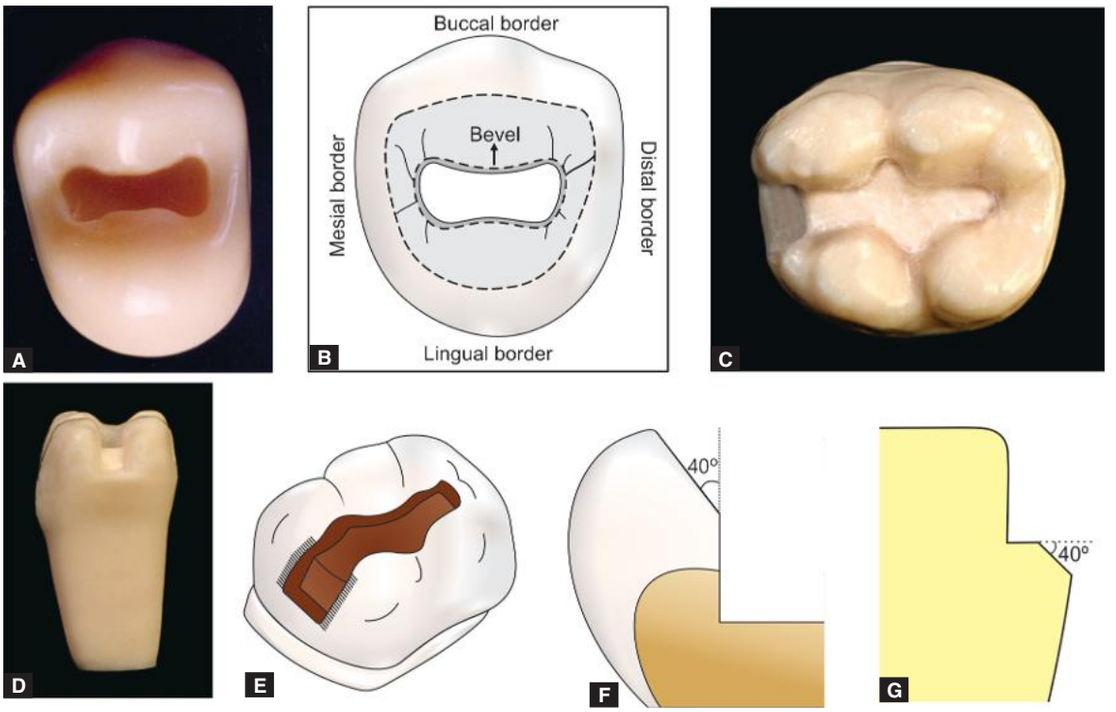


Fig. 16.4: Class I cavity preparation for cast gold (maxillary second premolar): **(A)** Occlusal view; **(B)** Diagrammatic; **(C)** Class II cavity preparation for cast gold (occlusal view); **(D)** Proximal view; **(E)** Diagrammatic representation (lines indicate secondary flare); **(F)** Occlusal cavosurface margin; **(G)** Gingival cavosurface margin

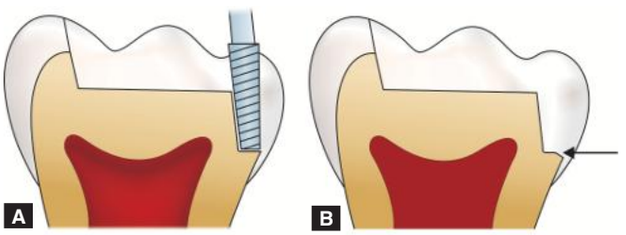


Fig. 16.5: Cervical wall in sound tooth tissue: **(A)** Finishing cervical wall; **(B)** Creating bevel at cervical cavosurface margin

Pulpal and cervical floors ideally should be perpendicular to the lines of force that will influence the restoration. Floors positioned perpendicular to these lines of force absorb the stress over a broad area of tooth.

Well-defined line angles are also important in obtaining resistance and retention form. These aid in maintaining precise relationship between restoration and tooth tissue.

The axiopulpal line angle is slightly rounded to dissipate the stresses (Fig. 16.6A).

Reverse bevel is placed at the gingival wall (Fig. 16.6B).

The slice is frequently used to increase the resistance and retention form by exposing a larger amount of tooth surface to the frictional grasp of the restoration (Fig. 16.6C, D).

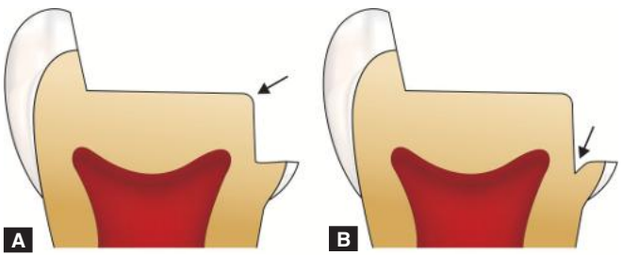


Fig. 16.6A and B: Axiopulpal and axiogingival line angle: **(A)** Rounded axiopulpal line angle (arrow); **(B)** Reverse bevel at the gingival wall (arrow).

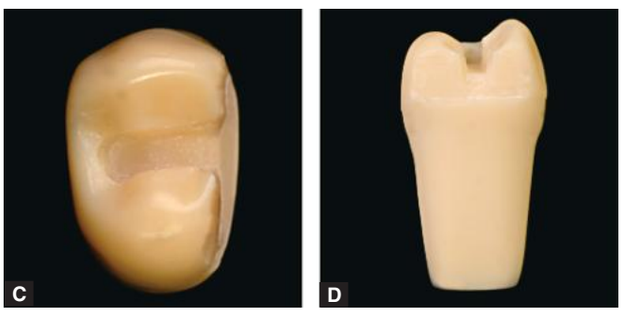


Fig. 16.6C and D: Slice preparation: **(C)** Occlusal view; **(D)** Proximal view

The occlusal interlock or dovetail is a major factor in resistance and retention form.

Tapered grooves, extending from cervical floor to the occlusal surface, are sometimes placed in the

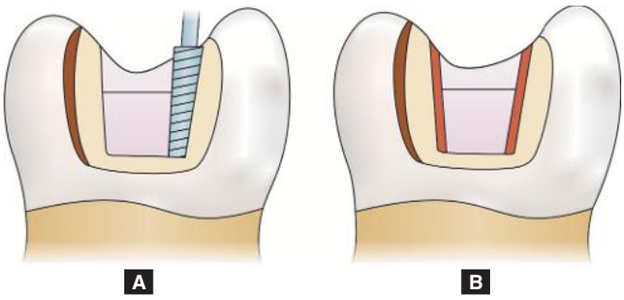


Fig. 16.7: Proximal grooves are incorporated to improve retention form: **(A)** Preparing grooves with tapering fissure bur; **(B)** Prepared grooves

dentin portion of the proximal walls to form a locking key to aid in preventing lateral dislodgement of the restoration (Fig. 16.7A, B).

Factors Affecting Retention of Cast Restorations

The factors affecting the retention of cast restorations are:

a. ***Magnitude of the dislodging forces:*** Generally, sticky foodstuffs act as a pulling force. The quantum of vertical and oblique forces also tends to dislodge the restoration. The magnitude of the dislodging forces depends on the stickiness of food, occluding and lateral movement of jaws, the surface area and texture of restoration being pulled.

b. ***Geometry of the tooth preparation:*** Retention of cast restorations depends on the geometric form of the preparation, rather than on the intervening cements. (Cements act by increasing the frictional resistance between tooth and the restoration.)
Cement is effective, only if the restoration has a single path of withdrawal, i.e. the tooth is prepared to restrain the free movement of the restoration. The geometrical forms generally responsible for retention are:

i. ***Taper:*** A basic requirement for the intracoronal cast gold restoration is that the cavity walls must diverge from the floor of the preparation externally. A similar requirement for all extracoronal preparations is that walls must converge from the cervical to the occlusal surface. This is the concept of taper. Removal of wax pattern and insertion of casting is facilitated by the taper.
Each intracoronal and extracoronal preparation has a line of draw that describes the path of insertion and removal of the casting, which determines the axis of taper (Figs 16.8A to C and 16.9A, B).
The axial length of the preparation will influence the amount of taper. Longer preparations require taper in the higher range; short preparations in the lower range. The degree of taper cannot be so great as to lose the frictional grasp between tooth and the restoration. It must, however, be sufficient to allow the complete seating of the restoration in the cavity.
Ideally, the tooth preparation should have parallel walls.
A slight taper is necessary for the fabrication of wax pattern. As long as this taper is small, the movement of the cemented restoration will be effectively restrained by limited path of withdrawal. As the taper increases, there will be free movement of the restoration along the path of withdrawal.
The maximum convergence between opposing walls should not be more than 6°. Preferably, two opposite surfaces must always be parallel in a cast restoration.

ii. ***Surface area:*** The retention of restoration is dependent on the length of path of withdrawal or more precisely on the surface area in sliding contact. Therefore, restorations with long axial walls are more retentive than those with short axial walls, e.g. restorations for molars are more retentive than for premolars. Surfaces where the restoration is essentially being pulled away rather than sliding, (e.g. occlusal surface), do not add much to total retention.

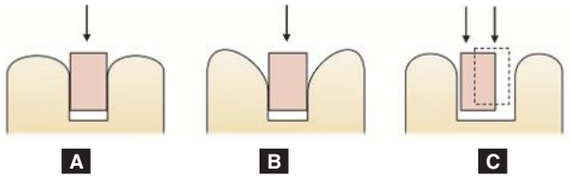


Fig. 16.8: Inlay taper: **(A)** Single path of insertion (arrow); wall height proper and parallel; **(B)** Single path of insertion (arrow); half of the wall height parallel; **(C)** Multiple paths of insertion (arrow) no retention, even if parallel walls with proper height

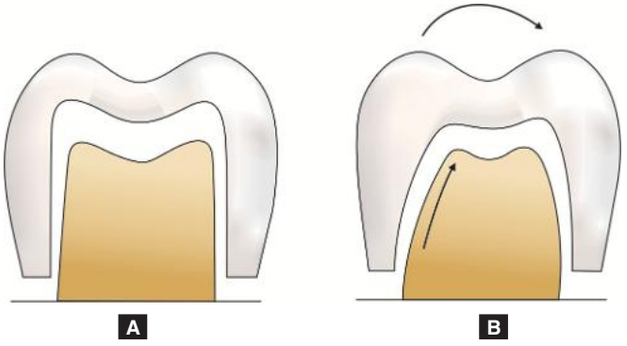


Fig. 16.9: Extracoronal taper: **(A)** Appropriate height and parallel walls provide better retention; **(B)** Appropriate height, but extra taper hinders retention (arrows showing movement of restoration under functional stresses)

iii. ***Stress concentration:*** It is established that the failure in retention form is always due to cohesive failure in the cement layer. The stresses are not uniform throughout the cement, but are concentrated around the junction of the axial and occlusal surfaces.

iv. ***Type of preparation:*** The retention of a complete crown is about double than that of partial coverage restoration.
Adding grooves to a preparation with a limited path of withdrawal does not markedly affect its retention, because the surface area is not increased significantly. However, where the addition of a groove limits the path of withdrawal, retention is increased.

c. ***Roughness of the surfaces being cemented:*** When the internal surface of a restoration is smooth, retentive failure occurs through the cement–restoration interface. The retention can be increased, if the inner surface of the restoration is roughened (inner surface is roughened with 50 µm alumina particles).

d. ***Materials being cemented:*** The more reactive the alloy, the more will be adhesion with the luting cement. (High reactive base metal alloys are better retained than less reactive gold alloys.)

e. ***Type of luting agent:*** The adhesive resin cements are most retentive as compared to other cements.

3. Convenience Form

Convenience form provides the accessibility and visibility required to complete operative procedures accurately. Extension, taper and flare of proximal walls to permit bevel placement, and extension to allow proper finishing are all examples of convenience form.

4. Removing Carious Dentin

Removal of deeper carious lesion precedes the establishment of resistance and retention form.

5. Finishing Enamel Walls and Margins

The marginal fit of a gold restoration depends upon the approximation of cast metal to tooth tissue. The surface (outer or inner walls) should be smooth; plain fissure burs can be used at the final stages of preparation.

The cervical bevel is placed with the flame shaped bur or gingival marginal trimmers (Fig. 16.10A, B). They provide a steeper bevel to prepare for an effective adaptation of the metal margin.

Whenever a direct wax pattern is to be formed, the bevel used is of greater bulk and extends a greater width across the cervical floor. Thus, a bevel on the cervical margin of a box preparation for the direct technique should be uniform, about ¼ to ½ of mesiodistal width of the cervical floor and must include the proximal cervical cavosurface angles.

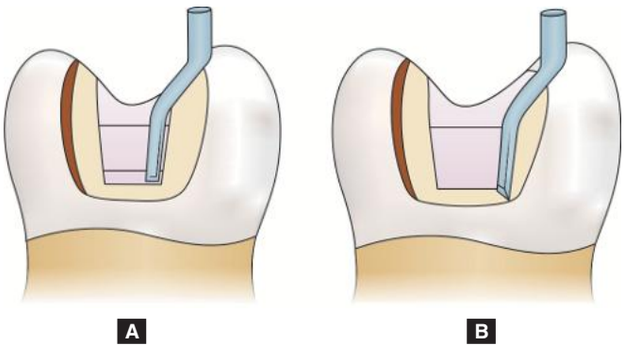


Fig. 16.10: Gingival marginal trimmer using for: **(A)** Finishing cervical wall; **(B)** Creating cervical carvosurface bevel

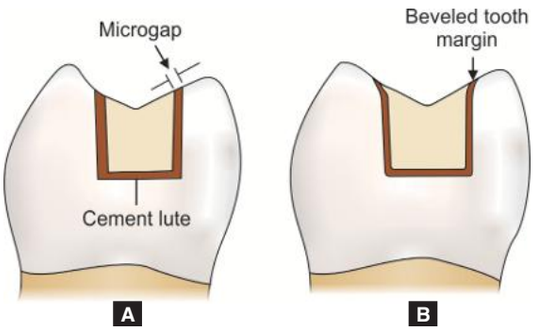


Fig. 16.11: Occlusal beveling of inlay: **(A)** Significant microgap is seen with a non-beveled inlay; **(B)** Microgap is reduced (virtually nil) with beveled inlay

The sharp angle formed at the occlusal outline should also be beveled (Figs 16.11A, B).

6. Cleaning (Toilet) the Cavity

The cavity preparation should be cleaned with distilled water/normal saline. After drying with cotton pledgets and a gentle stream of warm air, the cavity should be scrutinized carefully for any imperfections. A trial impression with gutta-percha (Fig. 16.12) is always helpful to evaluate taper and the line of draw of the preparation.

TYPES OF MARGINS IN A CAST RESTORATION

Various types of margins are tried in a cast restoration; however, bevel, chamfer and the shoulder are commonly used margins (Fig. 16.13 A to C). The difference between chamfer and shoulder is that chamfer is less than 0.5 mm wide and is rounded; while, shoulder is more than 0.5 mm wide and sharp. The chamfer is given where the metal is malleable; whereas, the shoulder is given where the metal is not malleable.



Fig. 16.12: Trial impression with gutta-percha

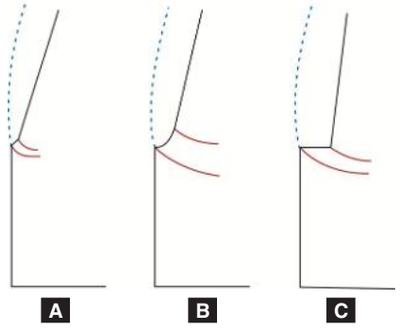


Fig. 16.13: Margin designs in a cast restoration: **(A)** Bevel; **(B)** Chamfer; **(C)** Shoulder

The relative advantages, disadvantages and indications of margin designs are summarized in Table 16.2.

Bevels

The weakest link in any cast restoration is the tooth–cement–cast interphase. An accurate wax pattern and reproduction in gold casting may not assure well-adapted margins to the tooth surface. This peripheral marginal anatomy of the preparation is called 'circumferential tie', created by placing bevels along the cavosurface margins.

The enamel must be supported by sound dentin, and enamel rods along the cavosurface margin should be continuous with sound dentin.

Bevels reduce the error factors (space between cast and tooth substances) at the margins compared to their internal dimensions, depending on the bevel's angulation. Bevels indirectly improve retention form for a cast restoration; bevels decrease the direct frictional component between the tooth and the casting.

The types of bevel are:

a. *Partial bevel:* It involves part of the enamel wall. Such type of preparation is indicated in direct filling gold restorations (Fig. 16.14A).
b. *Short bevel:* It involves the entire enamel wall. This type of bevel is preferred in cast gold restorations (Fig. 16.14B).
c. *Inverted bevel:* It is given on the labial shoulder of metal ceramic crowns to effectively improve the esthetics at the margins (Fig. 16.14C).
d. *Reverse bevel:* A reverse bevel is placed at the dentinal portion of the cervical wall towards the axiogingival line angle (Fig. 16.14D). The hydrostatic pressure during cementing a cast restoration can produce a rotational displacement of the casting with flat gingival walls. This effect is resisted by the reverse bevel resulting in even seating of the cast restoration.

Table 16.2: Margin designs: Relative advantages, disadvantages and indications

Preparation margin design	*Advantage*	*Disadvantage*	*Indications*
Bevel	• Removes unsupported enamel • Allows finishing of metal	• Extends preparation into sulcus, if used on apical margin	• Facial margin of maxillary metal ceramic restorations • Intracoronal cast restorations • Reverse bevel at gingival wall (gold inlays)
Chamfer	• Distinct margin • Easy to control	• Care needed to avoid unsupported lip of enamel	• Lingual margin of cast metal restorations and metal ceramic crowns
Shoulder	• Provide bulk of restorative material	• Less conservative • Difficult in mandibular incisors	• Labial margins of metal ceramic crowns • Cervical margins of all-ceramic crowns

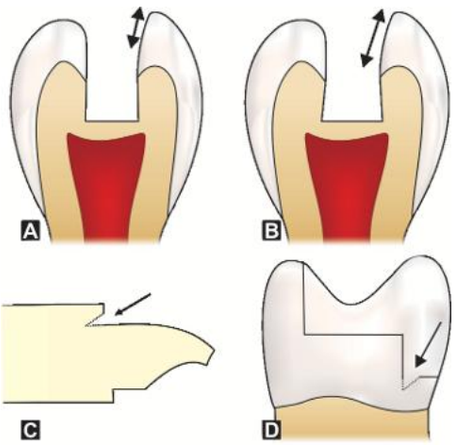


Fig. 16.14: Types of bevel: **(A)** Partial bevel; **(B)** Short bevel; **(C)** Inverted bevel; **(D)** Reverse bevel

FLARES IN CAST RESTORATIONS

The requisite flares in cast restorations are:

a. ***Primary flare:*** As per the old concept of inlay cavity preparation, the proximal walls were divided into two halves. The axial half was placed at 90° to the axial wall and proximal half had an angulation of 45° to the axial half. This proximal half that helped to bring the proximal wall into self-cleansing areas was known as the primary flare. It was believed that the axial wall should be at 90° to the proximal walls to provide the retention and resistance form to the cavity, but later it was found that flaring of the walls right from axiofacial or axiolingual line angles till the cavosurface margin did not alter the retention and resistance form. This implies that the flaring of lingual wall and buccal wall of proximal box from the axial wall (axiolingual and axiofacial line angles, respectively) to the proximal cavosurface margins is known as primary flare (Fig. 16.15A, B).

Functions: The function of the primary flare is to keep the proximal buccal and lingual walls in the self-cleansing areas.

b. ***Secondary flare:*** The secondary flare is a flat plane superimposed peripherally to a primary flare. It is as good as the occlusal bevel given to create a circumferential tie in gold inlays to take the advantage of malleability of gold alloys (Fig. 16.15C).

Functions: A secondary flare creates the obtuse angulation at the proximal cavosurface margin; created exclusively for gold inlay preparations (malleability of gold facilitates adaptation of restoration to proximal cavity walls).

The modified preparations, such as box only preparation, slice preparation and modified flare preparation, are no longer used.

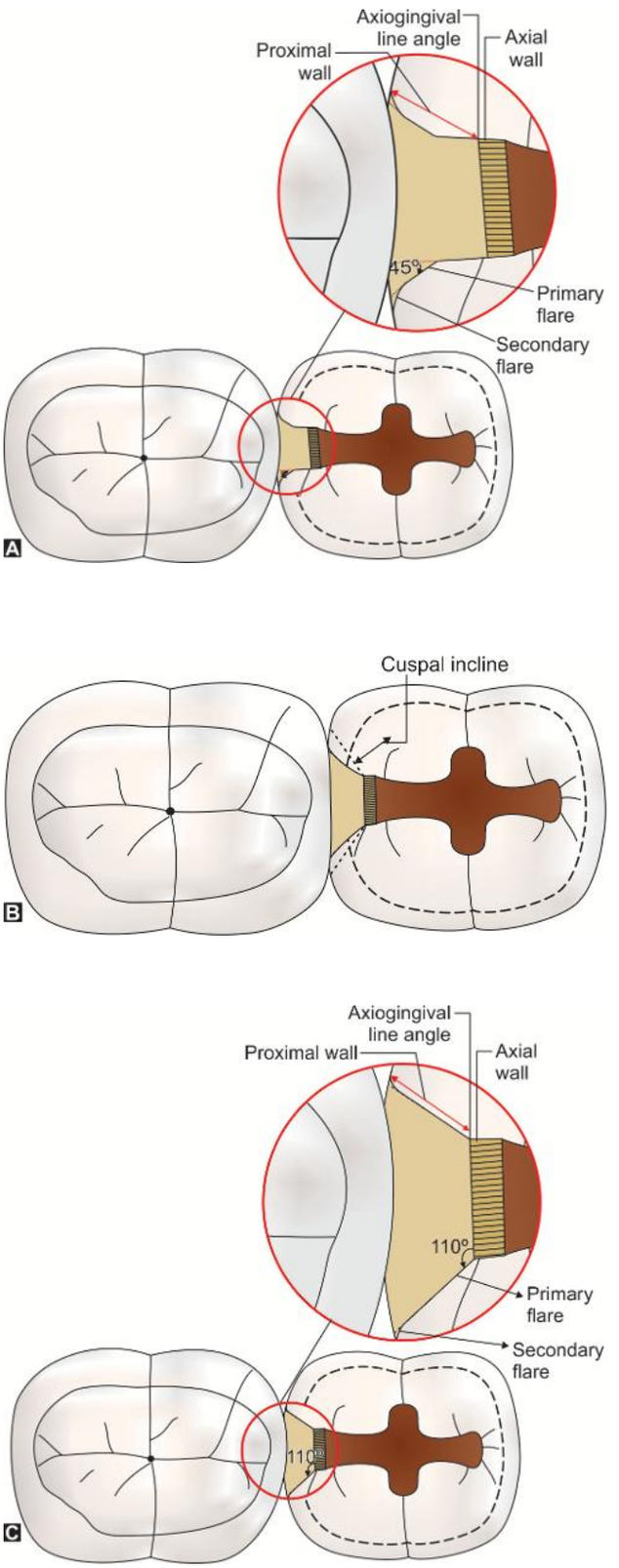


Fig. 16.15: Flares in inlay cavity preparation: **(A)** Old concept; **(B)** Present concept; **(C)** Extension of proximal wall in broad contact areas

CUSP CAPPING/ONLAY

Inlay vs Onlay

An inlay is primarily an intracoronal cast restoration that is designed mainly to restore occlusal and proximal surface(s) of posterior teeth without involving the cusps. An onlay is a combination of intracoronal and extracoronal cast restoration, when one or more cusps are covered. A full veneer crown is an extracoronal cast restoration, where all cusps are covered.

Cusp capping or onlays are usually indicated where one or more cusps, which are weakened by caries or trauma, are to be covered occlusally. The cast restorations are suitable especially when functional cusps are to be capped. It is partly an intracoronal and partly extracoronal type of restoration. If the cavity involves more than half of the cusp inclines, cuspal coverage is mandatory.

A tapering fissure bur of appropriate size (Fig. 16.16) is used to reduce the cusp occlusally by 1.0–1.5 mm.

The design of extracoronal margins, to be placed at the enamel, can be selected depending upon the restorative material (Fig. 16. 17A to C). Usually chamfer is indicated with cast gold restoration. Because of the ductility and malleability of the material, the margins can be burnished and adapted to the tooth surface.



Fig. 16.16: Different sizes and shapes of bur for cast preparation

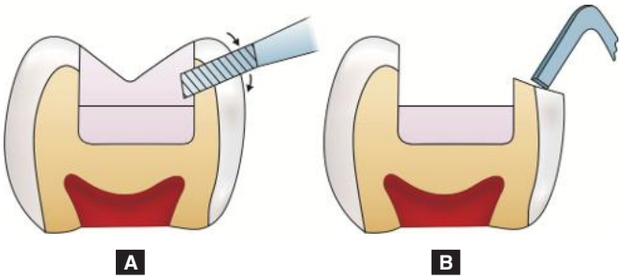


Fig. 16.17: Cusp capping/onlay: **(A)** Cusp being reduced with an appropriate bur; **(B)** Margin being refined with a hand instrument

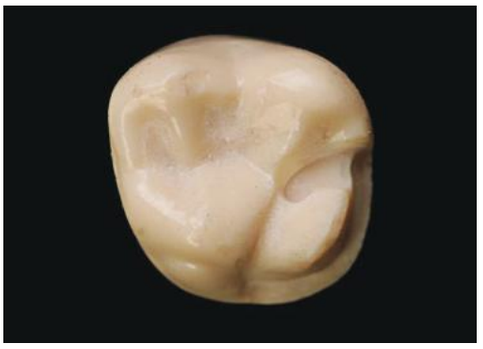


Fig. 16.17C: Class II cavity preparation with distolingual cusp reduction

In case, functional cusps are capped, it is advisable to make contra-retention on the opposing tooth surface. For example, if both the buccal cusps of mandibular molars are capped, lingual extension of the cavity is mandatory, as is done in lingual complex of Class I cavities.

Tucker's Technique

Dr. Richard Tucker developed a conservative cast gold inlay preparation technique. The cavity preparation sequence is as follows:

After placement of rubber dam, the existing restoration, if any, and carious tooth structure are removed.

The occlusal box is prepared with straight fissure bur and pulpal floor is placed to a depth of 1.5 to 2.0 mm. A taper of 3° to 5° is uniformly maintained. The proximal box form is established and blended with the occlusal preparation. The buccal and lingual walls must be extended sufficiently (1.0 mm) beyond the adjacent tooth. The proximal walls have slight flare (45°) that eliminate unsupported enamel rods. Occlusal and gingival bevels are placed and cavity walls are finished. The bevel should be definitive and smooth, but not too wide. Tucker's gingival margin trimmer is used in this technique. No. 232 Tucker (10-98-10-16) GMT is used on distal aspect and No. 233 Tucker (10-78-10-16) GMT is used on the mesial aspect. The bevels on non-Tucker GMT instruments are less acute (45°) than the ones marked Tucker (30°). The occlusal bevel is placed with the straight fissure bur inclined only a few degrees more than that of the occlusal wall. The finishing of the proximal walls is completed with a 0.5 inch garnet disc. The technique is illustrated in Fig. 16.18.

A thin film of calcium hydroxide is placed on the internal surface of the preparation. It acts as a separator for easy removal of composite build-up prior to cementation. The use of resin composite build up conserves tooth structure. The resin composite

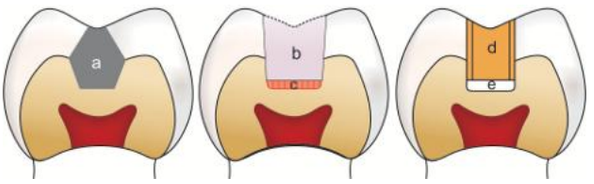


Fig. 16.18: Tucker's technique: **(a)** Carious lesion; **(b)** Cavity preparation for inlay; **(c)** Composite build-up; **(d)** Inlay; **(e)** Luting cement

build-up allows the operator to cut a precise cavity preparation that has an ideal taper, smoothness and proportions. An autocure resin composite can be placed with a packing instrument or a syringe. The build-up is removed and the casting is fabricated.

Differences in cavity preparation for silver amalgam and cast restorations: The differences in cavity preparation for silver amalgam and cast restoration are tabulated in Table 16.3.

FABRICATION OF CAST RESTORATIONS

Fabrication of cast restoration involves various procedures. These procedures are equally important as that of cavity preparation for the success of the treatment.

The procedures are as follows.

A. Impression Technique

In dentistry, impression is the negative replica of teeth and surrounding tissues. From this negative form, a positive reproduction or cast is made. The cast must exactly duplicate the preparation in order to have the precision fitting castings.

Several types of impression materials are available for use in the fabrication of cast restorations.

The impression can be taken on prefabricated tray or the tray can be fabricated outside the oral cavity.

Selection of Impression Tray

For reversible hydrocolloid, stock metal tray is used. For rubber base materials, stock or custom trays can be used (Figs 16.19 and 16.20).

Custom trays are usually preferred because they allow a more uniform, thin layer (2.0–3.0 mm) of the impression material. A uniform thickness of impression material will lead to less distortion.

Making Impression with Elastomeric Materials

The polyvinyl siloxane (addition silicones) is commonly used.

Local anesthesia is administered. Gingival retraction procedures are carried out. The retraction cord is packed into the sulcus for displacement of the gingival sulcus to provide access to subgingival margins of the preparation.

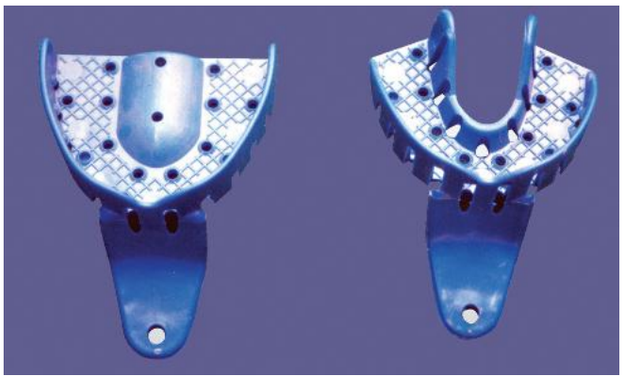


Fig. 16.19: Stock impression trays



Fig. 16.20: Custom impression trays

Most silicones are available in the form of two pastes (base and catalyst) that are mixed prior to application on the tooth.

The impression material is injected into the sulcus after removing the cord. The tray with impression material is seated firmly in one position. The impression tray is held for 6–8 minutes in place.

The impression is removed as quickly as possible and in vertical direction. It is rinsed, dried and disinfected using 2.0% glutaraldehyde for 10 minutes and is then ready for pouring (Fig. 16.21).

B. Construction of Die from the Cast

A 'die' is the positive replica of one tooth. Usually the prepared tooth is separated from the cast (die making) and the rest of the procedures are carried out on the die (Figs 16.22 and 16.23).

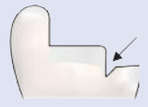
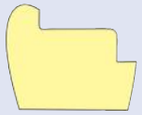
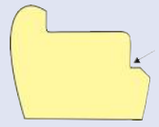
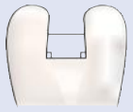
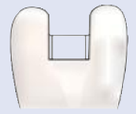
Materials for Die Construction

Various materials have been used for construction of dies.

The *basic requirements* for such materials are:

- Compatible with the impression material.
- Dimensionally stable.
- Should have a smooth, non-abradable surface.

Table 16.3: Differences in cavity preparation for silver amalgam and cast restorations

Silver amalgam	*Cast restorations*
i. Intercuspal width is 1/4th intercuspal distance (outline form is narrow). 	Intercuspal width is 1/3rd intercuspal distance (outline form is wide). 
ii. Cavity depth is more. 	Comparatively less depth. 
iii. Cavity walls are kept convergent occlusally (minor undercuts). 	Cavity walls are kept parallel; 2°–5° divergence occlusally (no undercuts) preferred. 
iv. Buccal and lingual proximal walls are convergent occlusally. 	Buccal and lingual proximal walls are parallel. 
v. No secondary flare 	Secondary flare indicated 
vi. Cavosurface bevel is contraindicated (butt joint is preferred). 	Cavosurface bevel (30–40°) is indicated. 
vii. All line angles and point angles are rounded; axiopulpal line angle is beveled. 	All line angles and point angles are well defined; axiopulpal line angle is slightly rounded. 
viii. No reverse bevel is given. 	• Reverse bevel at gingival wall. 
ix. Minor gingival cavosurface bevel (removing unsupported enamel) 	• 30°–40° gingival cavosurface bevel 
x. Grooves are not given; only locks are given. 	Grooves are given; locks are not given. 

- Able to accommodate auxiliary retention devices (e.g. pins).
- Should be able to receive a spacer in selected areas to create space for the luting agent of the casting.

The materials used for construction of dies are:

a. ***Gypsum:*** The most commonly used die materials are type IV and type V dental stone.

Strength of these materials is greatly increased by decreasing the porosity; pouring under vacuum is helpful. Decreasing the water/powder ratio increases the strength.

b. *Electroformed dies:* Certain metals having high strength, hardness and abrasion resistance can be electroplated on the impression forming electroformed dies.

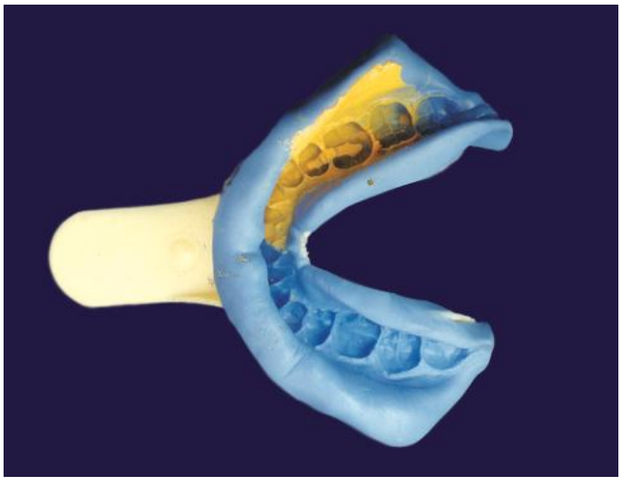


Fig. 16.21: Elastomeric impression ready for pouring



Fig. 16.22: Die cutter

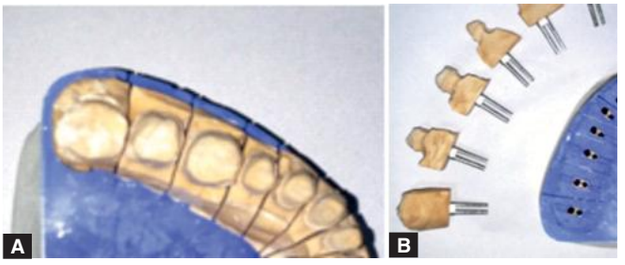


Fig. 16.23: Die preparation: **(A)** Trimmed die; **(B)** Removal of die from the cast

Electroplating is the result of electrolysis. When electric field is superimposed on the electrolytic solution, the ions in this solution are attracted and begin to move to oppositely charged electrodes. The positively charged ions (cations) move to the negative electrodes (cathodes), the negatively charged ions (anions) move to the positive electrode (anode).

The anode is a bar of pure metal, supplying metal cations continuously. The deposition takes place on the cathode (impression).

The materials used for electroplating are:

i. Copper plating
ii. Silver plating
iii. Silver amalgam
iv. Silicophosphate cement dies
v. Epoxy resin dies

C. Preparing the Wax Pattern

Two methods are generally used to prepare the wax pattern—the direct method (on the tooth) and the indirect method (on a die).

a. *Direct Method*

Indications

- A tooth is in an area of easy accessibility, i.e. the tooth should be clearly visible and easy to work on (second and third molars are usually not suitable).
- Cavity preparations with minimal proximal extensions, i.e. relatively small cavity preparations.
- Cavity preparation where the walls are flat, internal line angles are sharp and gingival bevel is definite.

Manipulation

The stick of inlay wax can be softened in hot water or using dry heat. (Softening in hot water can result in inclusion of droplets of water that would splatter on flaming).

When wax is softened directly over a flame, care must be exercised to ensure thorough heating of the wax stick. The wax should be kept moving till it becomes shiny and then removed from the flame. It should then be compressed between the fingers and again warmed. The process is repeated until the wax is warm throughout. It is kneaded using the thumb and first finger, which are lubricated with vaseline to avoid sticking of the wax.

A wax annealer, which maintains a constant temperature of 65°C, can also be employed.

The pattern should be prepared in such a manner that no addition of wax would be required, as wax, which is added after the initial cooling, introduces stresses, which will distort the pattern.

The wax pattern by direct method can be prepared with and without the application of matrix band.

i. ***Wax pattern prepared with a matrix band:*** The retainer and band are fitted on the tooth, making certain that the gingival margins are covered by the band before applying a wedge. The internal

surface of the band is lightly lubricated with a separating medium, such as castor oil. The softened wax is inserted into the cavity and pressed with finger.

Finger pressure is maintained while the wax is cooled and hardened. The bulk of excess wax is trimmed (Fig. 16.24). The matrix retainer is loosened and removed. Trial removal of the pattern is attempted at this stage.

The pattern is carefully inspected for reproduction of sharp internal details and the cavosurface/gingival margins.

If not satisfied, repeat the process (repair of pattern is not advisable).

The pattern is again seated in the cavity.

The pattern is carved with the help of plastic instruments. To evaluate the occlusion, the tooth is carefully isolated and dried with cotton rolls. The opposing tooth is also dried. With a camel hair brush, talcum powder is dusted on the occlusal surface of the wax and the patient is instructed to close the teeth lightly. Cuspal contact with the wax pattern is indicated by a shiny burnished spot on the pattern. The spots of hyperocclusion are carved away using warm carvers.

The occlusal surface may be polished by rubbing a wet pledget of cotton over the area. The pledget of cotton can be warmed over a flame for better results. Proximal surfaces are polished by wrapping a smooth linen strip tightly around the tooth and clinching it upon the buccal side.

Advantages

- The pattern is carved on the tooth and not on a model. (Model may not be a perfect replica of the tooth because of possible inaccuracies during each stage of preparation of the model.)
- Laboratory work is decreased.
- Time saving (though chairside time is increased, but the overall time required for fabrication is decreased).

Disadvantages

- Great skill and patience is required to carve patterns in the oral cavity.
- When wax is carved by indirect vision in a mirror, manipulation becomes difficult.
- Discrepancies of the pattern at the gingival margin are difficult to detect until the pattern has been withdrawn.
- Valuable chairside time is lost.
- If the casting fails, the patient has to be recalled; wasting time and energy.



Fig. 16.24: Wax pattern fabricated with band applied

ii. ***Wax pattern prepared without a matrix band:*** The inlay wax stick is softened to form a pointed end. The softened pointed end is forced into the cavity, the harder end acting as a plunger so as to confine the wax to the approximal part of the cavity; the thumb and forefinger of the free hand are pressed into the buccal and lingual embrasures. Excess wax is trimmed off and burnisher is used to press the wax further into the cavity.

The patient is asked to close his teeth to force the wax further into the cavity. The occlusal surface is carved with a wax carver. Wax in the embrasures is trimmed.

The approximal surface is burnished with a smooth linen strip. The strip should be held in such a way so that it does not burnish the wax away from the cavity margin (Fig. 16.25).

The proximal contour and contact of the pattern deserves special attention. It is important that contacts be of proper form as in the right position.



Fig. 16.25: Wax pattern fabrication without band applied on tooth



Fig. 16.26: Method to measure contact



Fig. 16.27: Die spacer and thinner

In order to evaluate, the contact form the ligature (dental floss) is first passed to the gingival area of the contact; the two ends are then held parallel in the occlusal direction as shown for the contact between the premolars (Fig. 16.26). This measures the faciolingual width of the contact of the two surfaces.

Without removing the ligature, the two ends should be held parallel in the facial direction as shown for the contact between two premolars. This measures the occlusogingival width of the contact. The contact is adjusted, if not proper.

b. *Indirect Method*

After the preparation of the die, a lubricant is applied to facilitate the withdrawal of the pattern. The time for the application of the lubricant depends upon the wetness of the die; the humid die will take less time and vice versa.

In case of large patterns, mostly with crown and bridges, die spacer is used to compensate the solidification shrinkage of the casting alloy (Fig. 16.27).

The blue inlay wax is softened, molded and pushed into the cavity as in direct pattern technique. The rest of the finishing and carving is carried out as in direct pattern.

D. Removing the Pattern from the Cavity

The pattern must be removed without distortion and maintaining the path of removal parallel with the direction of the cavity walls. The sprue former should be attached to the pattern while it is still on the tooth to minimize distortion of pattern due to heat and mechanical induction of stresses. The sprue former of exact size and shape is selected, keeping in mind the size of wax pattern, casting machine to be used and the metal to be cast. Sticky wax is applied to one end of the sprue and attached to the pattern site. A little wax can be added, at the joining of the sprue and the pattern. After keeping the sprue for 2–3 minutes, trial removal can be carried out. Indirect procedure can also be used. A copper staple or a 24 gauge twisted brass wire is inserted into the occlusal part of the pattern. For MOD patterns, the prongs of the wire are placed near the marginal ridges (Fig. 16.28) and for disto-occlusal or mesio-occlusal patterns, the prongs are inserted at an angle of 45° near the marginal ridge (Fig. 16.29). Using the staple, the pattern is lifted out of the cavity with a direct pull parallel to the cavity walls. The sprue is then inserted into the wax pattern and the staple removed by holding it with warm pliers to melt the wax holding the staple (Fig. 16.30).

Low fusing inlay wax is flown over the contact area and flushed with the proximal surface to build the contact (Fig. 16.31).

E. Sprue Former

i. *Sprue Former Material*

The sprue former can be made of wax, resin or metal. Wax and resin sprue formers have the advantage of

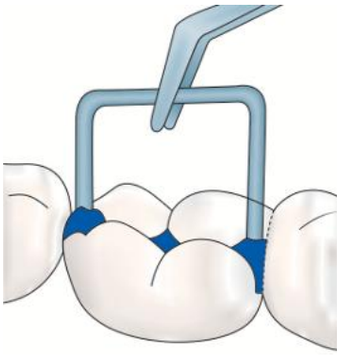


Fig. 16.28: Copper staple for removing wax impression of MOD cavity

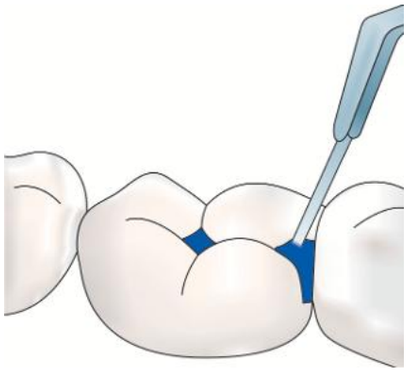


Fig. 16.29: Staple inserted at 45° for mesio-occlusal or disto-occlusal preparation



Fig. 16.30: Removal of pattern from the tooth

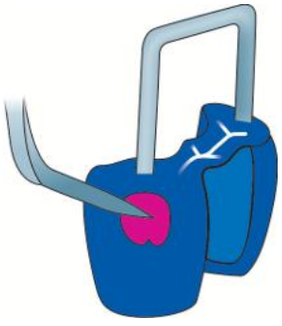


Fig. 16.31: Building the contact area with low fusing wax

being burnable and so do not need to be mechanically removed (Fig. 16.32). Further, the wax or resin has a fusion temperature lower or almost same as that of the pattern wax. However, wax sprue formers lack rigidity; a major disadvantage.

Metal sprue formers can be solid or hollow. Hollow sprue formers are preferred, since they hold less heat than a solid sprue former and transfer less heat to the wax pattern resulting in less distortion. Also, their retention to the wax pattern is better.



Fig. 16.32: Inlay wax for making sprue former

The metal sprue former must be mechanically removed prior to burnout. This could cause investment to loosen from the walls. To avoid this, metal sprue formers are uniformly coated with wax before investing, so that at the time of burn out, the sprue former comes out on its own because of melting of wax (the loosening breaking of investment can be avoided).

ii. *Sprue Former Diameter*

The diameter of sprue former depends on the size of wax pattern, quality of casting machine, and the ring, used to form the mold.

Preferably, the diameter of the sprue former should not be more than 1/4th of the total area of the wax pattern.

The diameter of the sprue is the most important factor in determining the speed with which the molten metal enters and fills the mold. Melt velocity is directly proportional to sprue diameter.

iii. *Sprue Former Length*

The length of the sprue former depends upon the length of the casting ring. The length should be adjusted, so that the wax pattern is within 6.5 mm (1/4 inch) of the open end of the ring for gypsum-bonded investments and 3.25 mm (1/8 inch) for phosphate-bonded investments. If the pattern is too close to the end of the ring, the molten alloy may blast through the investment during casting; if it is too far away, gases may not escape rapidly enough to allow metal to fill the mold space.

iv. *Location of Sprue Former*

The sprue former is attached to the bulkiest part of the wax pattern; the reasons are:

- Minimize the effect of released residual stresses by heat of attaching the sprue.

- Ensure that the thinner cross-section of the mold will be filled completely.
- Melt will always be fluid and available until all sections are adequately filled.

The general rule is to attach the sprue former in a position so that the molten metal will reach the mold area farthest from the attachment of the sprue former at the same time.

v. *Angulation of the Sprue Former*

Sprue former should not be attached at right angle to a broad surface because the melt may impinge the mold surface and produce a 'hot spot', creating suck back porosity.

An angle of 45° is usually adequate. The sprue former should be directed away from any thin or delicate parts of the pattern, since molten metal may abrade or fracture investment in this area and result in casting failure.

vi. *Number of Sprues*

If the wax pattern is designed such that attachment of the sprue former at the bulkiest portion allows the metal to flow uninterrupted, then a single sprue is adequate.

If multiple sprues are used, they should join together at the crucible former level in a reservoir larger in diameter than all the sprues combined.

vii. *Sprue Former Attachment*

The sprue pattern joint must be smooth and uninterrupted. If high velocity ingress of the melt into the mold is required, the junction should be flared.

To minimize stress release while joining a sprue former to a wax pattern, a drop of sticky wax should be applied to the wax pattern.

viii. *Reservoir*

In case, the diameter of the sprue is smaller than the average cross-sectional area of the pattern, a reservoir is attached with the sprue as near as possible to the pattern sprue junction. The diameter of the reservoir should be more than the average cross-sectional area of the pattern. The rationale of reservoir is to provide molten metal to the pattern.

When molten metal strikes against the mold wall at 90° angle and if the mold-metal temperature differential is more, it creates a hot-spot, the molten metal in this area will solidify last, leading to localized shrinkage porosity. The reservoir provides the molten metal to compensate for this localized shrinkage porosity.

If a sprue former, the diameter of which is greater than the diameter of the cross-sectional area of the pattern, is used, there is no need for providing the reservoir. The sprue will itself provide the molten metal.

Table 16.4 summarizes principles of sprue former.

F. Attaching Pattern to Crucible

The sprue is attached to the crucible in the same way as the sprue is attached to the mold (bulkiest in cross-section, flared and smooth).

The depth of the crucible is dictated by factors similar to those governing the diameter of sprue, i.e. alloy density, casting machine energy, melt viscosity, size of the pattern, porosity of investment, etc. The deeper the crucible, more velocity will be imparted to the melt on its way to the mold.

G. Preparation of the Ring

The metal used in the construction of a ring should be non-corrodible, hard and with a thermal expansion similar to that of investment used. Stainless steel is the most acceptable.

The ideal dimension of the ring should be 29 mm in diameter and 38 mm in height; however, it can vary according to size of the pattern.

Need for Liner

A resilient liner is placed inside the ring to provide a buffer against which the investment can expand

Table 16.4: Principles of sprue former

1. Sprue former length	• Depends on casting ring, such that 6.0 mm free at the trailing end
2. Sprue former diameter	• Equal to thickest part of wax pattern
3. Sprue former attachment	• Flared for gold alloy; reservoir is attached as near the attachment as possible
4. Sprue former material	• Wax, plastic or metal
5. Sprue former type	• Hollow or solid
6. Sprue former location/position	• Point of greatest bulk in the pattern to avoid distorting thin areas
7. Sprue former angle	• Attached at 45° to proximal area • In MOD preparation, λ (reverse Y) type attachment

to enlarge the mold. If there is no liner present, the investment is in direct contact with the walls of the mold and will not be able to expand outward; may expand towards the center of the mold, resulting in distortion of the casting. The liner, if wetted, allows for semihygroscopic expansion of the investment. It becomes easier to remove the investment and casting from the ring, if a liner is used.

Liner Material

For many years, asbestos was used to line casting rings; however, its use has decreased because of concern over its carcinogenic properties.

Presently, absorbent cellulose and non-absorbent ceramic materials are used. Cellulose liners readily absorb water when immersed and, therefore, must be wetted before use. Ceramic liners are virtually non-absorbent and can be used dry.

How to use a Liner?

A liner can be used dry or wet. A dry liner will allow greater normal setting expansion in the investment.

Theoretically, a wet liner will allow greater normal setting expansion and semihygroscopic expansion, but it also reduces the powder-water ratio, which in turn will reduce the thermal expansion of the investment. As a result, the net expansion with a dry liner will be slightly greater than with a wet liner.

The maximum thickness of a liner is 1.0 mm. A thicker liner or two layers of liner can also be used.

Liner should be placed 1/8 inch short at the ends to get less distortion of the mold.

After the liner has been placed in the ring, the ring is dipped in water for 10 seconds (in case of asbestos or cellulose liners) and then gently shaken to remove excess. Squeezing the liner would result in removal of excess water. The diagrammatic representation of the investment ring, explaining crucible former, sprue former, reservoir and wax pattern is shown in Fig. 16.33.

INVESTMENT OF THE PATTERN

a. Preparing the Wax Pattern for Investment

The wax pattern should be cleaned of any debris, grease or oils before it is positioned in the ring. This will decrease the surface tension and improve the wettability of the wax pattern.

Gentle washing with liquid soap using no. 2 paint brush is effective. The soap should be thoroughly rinsed off with water at room temperature and the pattern dried with a stream of clean air.

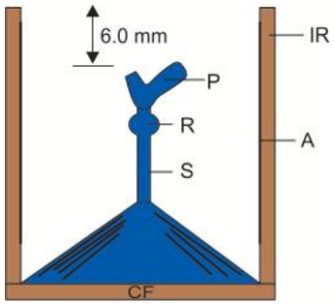


Fig. 16.33: The length of the sprue is so adjusted to bring the wax pattern (P) to 6.0 mm from the top of the ring. Crucible former (CF), investment ring (IR) lined with wet asbestos (A) kept short of the ring on both sides and reservoir (R) added to the sprue (S).

b. Investment Materials

The procedure for forming the mold is described as investing. These materials can withstand high temperatures; also known as refractory materials.

There are three types of investment materials:

i. Gypsum-bonded investments
ii. Phosphate-bonded investments
iii. Silica-bonded investments

i. *Gypsum-Bonded Investment*

Composition

- Refractory—silica (quartz or cristobalite) 60–65%
- Binder—alpha-hemihydrate 30–35%
- Modifiers—carbon, boric acid or sodium chloride 5%

Properties

- When gypsum is heated to high temperature (700°C), it shows slight expansion followed by ample amount of contraction.
- When heated, quartz changes its crystalline form (the alpha form converted into beta form), leading to increase in volume and rapid increase in linear expansion.
- A mixture of silica and dental stone results in greater setting expansion than gypsum product alone.
- When gypsum products are allowed to set in contact with water, the amount of expansion exhibited is much greater than normal setting expansion.

ii. *Phosphate-Bonded Investment*

The metal ceramic alloys and cobalt chromium alloys have high melting temperatures. They are cast in molds at 850°C to 1100°C. At high temperatures, gypsum-bonded investment disintegrates. Hence, investment, which can withstand higher temperature, is required, viz. phosphate bonded and silica bonded.

Composition

- Powder [refractory—silica, binder—ammonium diacid phosphate, magnesium oxide (reacts with phosphate ions liberated by ammonium diacid phosphate)]
- Liquid (silica solution in water)

iii. *Silica-Bonded Investment*

In silica-bonded investment, the silica is used as binder. It is derived from ethyl silicate or aqueous dispersion of colloidal silica or sodium silicate.

c. Methods of Investment

The methods of investment commonly used are:

i. Manual investing
ii. Vacuum investing

i. ***Manual investing:*** Water and powder, incorporated in the appropriate ratio (indicated by manufacturer) in a rubber bowl, is spatulated with a hand spatula. The mix is placed on a vibrating table and stirred slowly for 30 seconds to remove any entrapped air. The investment is carefully applied to the pattern, using a small sable brush. The investment is allowed to set partially. The rest of the lined ring is then filled with investment.

ii. ***Vacuum investing:*** The investment is mixed and the pattern is invested under vacuum. Several types of vacuum investing equipment are available in the market .

CASTING THE PATTERN

The investment contracts as it cools because of the low thermal conductivity of the investment. After waiting for 40–60 minutes, the temperature can be raised.

Three types of casting machines are utilized:

i. Air pressure casting machine (machines exerting pressure on the gold; by air, nitrous oxide gas or steam).
ii. Vacuum casting machine (machines drawing the gold into the mold by a vacuum).
iii. Centrifugal casting machine (revolving machines; gold is carried into the mold by centrifugal force).

i. ***Air pressure casting machine:*** The alloy is melted in the hollow left by crucible former and then air pressure is applied through a piston, which is pushed downward into contact with the top of the ring, enclosing the molten gold alloy. A pressure of 10–15 psi is usually applied (Fig. 16.34).

ii. ***Vacuum casting machine:*** A vacuum is applied through the base beneath the casting ring and the molten alloy can be drawn into the mold by suction. Mostly, combination of centrifugal pressure and a vacuum is used to create the driving force.



Fig. 16.34: Air pressure casting machine

iii. ***Centrifugal casting machine:*** The machine (Figs 16.35 and 16.36) has a spring, which can be wound into tension by rotating the arms with the weights at one end and the casting ring at the other. In front of the ring is a separate crucible in which the gold alloy is melted. When the spring is released, the two arms rotate rapidly, and the molten metal is forced into the mold by centrifugal force. A pressure gradient creates pressure of 30–40 psi at the tip of the casting. The gradient of heat transfer is such that the heat transfer is maximum at higher pressure end, ensuring that the tip solidifies first.

Time required to cast gold alloy by centrifugal force depends on the cross-sectional area of the sprue and the number of winds of the machine.

When the size of the sprue is increased, the casting time is reduced. When the number of winds is increased, increasing the force, the casting time is reduced.



Fig. 16.35: Induction (vacuum) casting machine

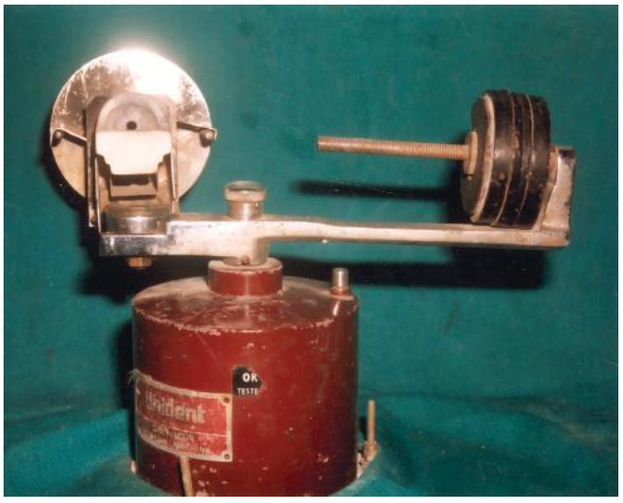


Fig. 16.36: Centrifugal casting machine

Procedure of Casting (Centrifugal Machine)

The spring is adjusted so that the base rests in front of the crucible. The spring is wound 3–5 times, depending upon the size of the pattern (the need of pressure).

Casting alloy is placed in the crucible. Enough bulk of the metal must be used in casting. The metal is melted using oxygen gas along with blow-pipe.

A conical flame, about 40 mm long, is obtained by adjusting the torch. The parts of the flame are identified (Fig. 16.37).

The reducing zone of the flame should be used to melt the metal.

A small amount of flux should be sprinkled onto the warm metal. Borax, when used as a flux, will help to exclude oxygen from the surface of the alloy and dissolve any oxides that are formed.

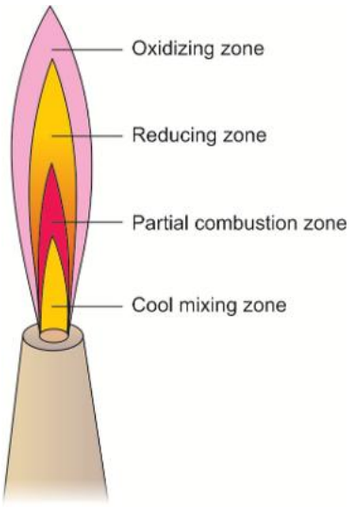


Fig. 16.37: Different zones of the flame

Reducing flux, which contains carbon in addition to borax, will reduce any oxides that may form. As the alloy melts, it appears spongy, and finally it assumes a spheroidal shape and moves with the flame.

Keeping the flame on the gold, the casting ring is removed from the oven and carefully placed in the cradle of the casting machine.

When the metal is light orange in color and tends to spin or follow the flame when moved, it is ready to be cast.

Gentle clockwise pressure is applied, the weight is released allowing the machine to spin. The metal will be thrust into the mold space.

The commonly employed casting techniques are as follows.

a. *Thermal Expansion (High Heat) Casting Technique*

The crucible former is removed after setting of the investment. The metal sprue former is warmed slightly over a gas flame and carefully removed. The inverted ring is tapped lightly on the laboratory bench to remove any loose particles, which may have fallen into the mold space.

If the burnout and casting procedures are to be delayed for several hours, it is advisable to place the invested pattern in humid environment to prevent excessive drying. Excessive drying of the investment prior to burnout may lead to molten wax getting absorbed into the investment.

The casting ring is placed in an oven preheated to approximately 900°F (480°C), held at that temperature for 20 minutes, and then the temperature is slowly raised to 1290°F (700°C) and held for 30 minutes.

Care should be taken to avoid heating gypsum-bonded investment above 1290°F. This is because above 1290°F (700°C) calcium sulfate is reduced by carbon molecules releasing sulfur dioxide gas. The sulfur dioxide gas, thus released, may contaminate the gold alloy as it enters the mold.

It is advisable to burnout with the sprue hole facing downward, since this will allow the wax to trickle down from the mold.

The methods used to melt gold alloy are as follows.

i. *Using a torch flame*
   - Natural gas (mainly propane) and air (lowest temperature; efficient for small inlays)
   - Natural gas and oxygen (higher temperature; used for extensive alloys).
   - Acetylene and oxygen mixture (hottest temperature; used for melting base metal alloys).

ii. *Using electric energy:* Induction heat is the most efficient method of melting an alloy. The metal is melted by an induction field that develops within

a crucible surrounded by water-cooled metal tubing.

b. *Hygroscopic Expansion (Low Heat) Technique*

This technique involves the immersion of the metal ring with investment into a warm water bath set at 37°C or 100°F.

The technique compensates shrinkage by three mechanisms:

i. 37°C water bath expands the wax pattern.
ii. Water entering the investment provides greater volume into which gypsum crystals can grow and provide some hygroscopic expansion.
iii. Thermal expansion at 468°C (burnout temperature for hygroscopic setting expansion).

The amount of hygroscopic expansion can be controlled by:

- *Water/powder ratio:* The higher the water/powder ratio, the lower the amount of expansion.
- *Time of immersion in the water bath:* The longer the delay before the investment is immersed in the water bath, the less is the hygroscopic expansion.
- Controlling the amount of water added during the setting.

Hygroscopic expansion can be achieved by two ways:

i. ***Water bath technique:*** A definite proportions of water and investment are used. Investment must be slow setting to allow sufficient time to expand. The wax pattern is invested. Both ends of the ring are sealed and it is placed in the water bath at 45°C until the investment has completely set. The wax is then burned out and the metal is casted.

ii. ***Controlled water technique:*** The amount of hygroscopic expansion is directly related to the amount of water available to the setting investment; adding a specific amount of water to the setting investment, a specific amount of expansion results. The controlled water technique is based on this principle.

For this technique, an investment, especially compounded to provide high hygroscopic expansion, is used.

The investment is mixed and poured into the rubber ring with supporting metal ring in place. Immediately after investing, the metal ring is removed so as to avoid restriction of the expansion of investment. The investment is flushed with top of the rubber ring and the reservoir collar inserted. The syringe is filled with water and the water is carefully dispersed into the reservoir collar. Generally, 1.1 mL water is added; however, less water (0.6 mL) can be added, if it is warm.

The investment is allowed to set; after setting, the rubber ring is removed and the investment mold without the support of a metal ring is subjected to burnout and the alloy is cast into the mold.

In the hygroscopic technique, the ring is heated around 460°C (thermal expansion of the investment is not desired).

Cleaning the Casting

After the casting has been completed, the ring is removed and quenched in water.

Advantages of quenching are:

- The noble metal alloy is left in an annealed condition for burnishing, polishing, etc.
- When water contacts the investment, it undergoes vaporizing; subsequently, produce cracking of the investment into small pieces. This simplifies cleaning the casting.

The fabrication of inlay is summarized in Flowchart 16.1.

FINISHING AND POLISHING

After casting, the gold alloys get discolored due to contamination/presence of sulfide in the investment material. Pickling procedure is carried out to remove the contaminants. The solutions used for pickling is 50% hydrochloric acid. Solution of sulfuric acid is considered advantageous. The best method of pickling is to place the casting in a test tube and pour acid over it. It is preferred to heat acid, but boiling should be avoided. After pickling, the acid is poured off and the casting is removed. The pickling solution should be renewed frequently, because it is likely to get contaminated. Casting should not be held with steel instrument, as this may contaminate the pickling solution and casting.

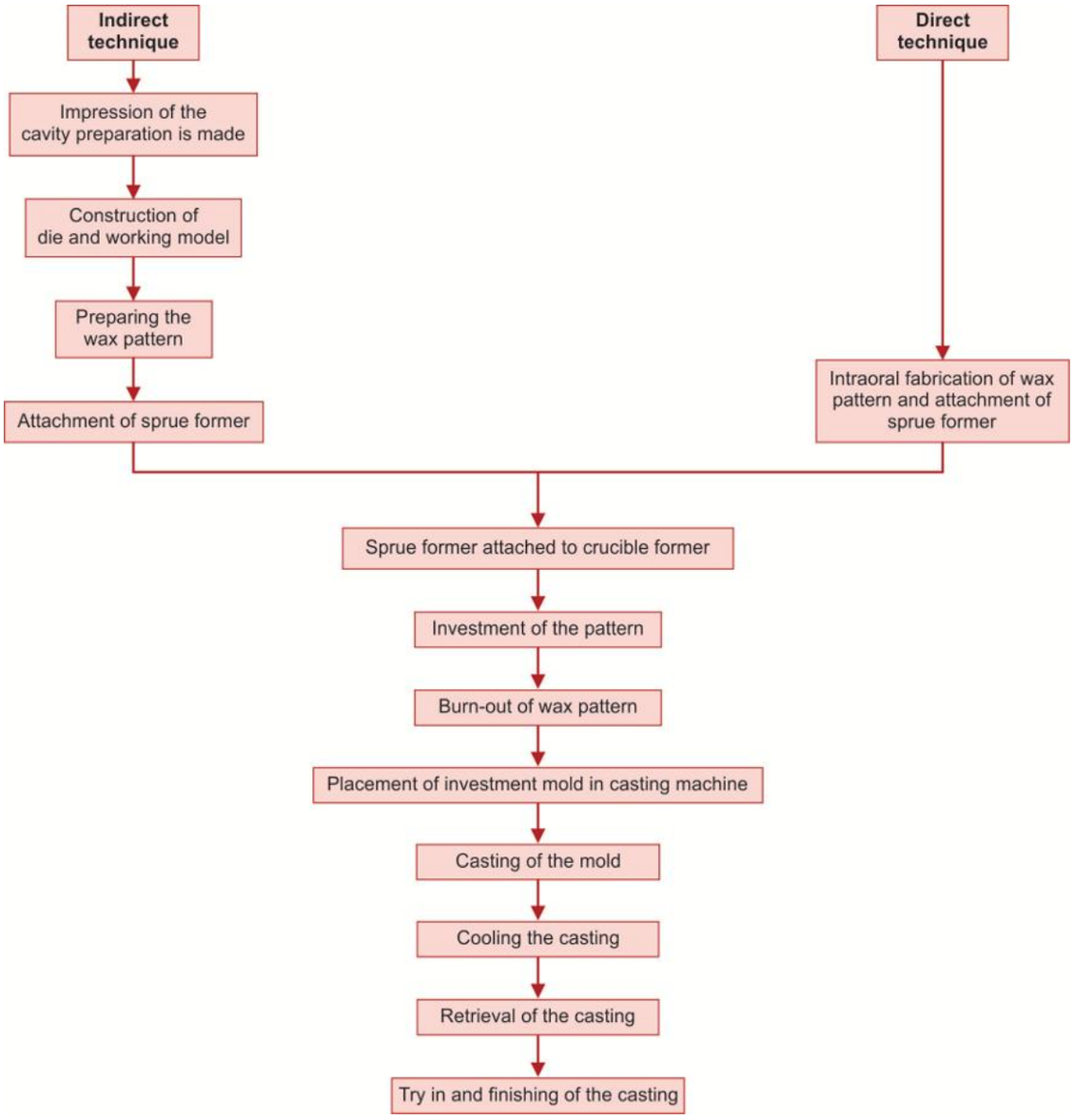
Another procedure to be avoided is heating the casting and then dropping it in the pickling solution. This may melt delicate margins of the casting or the casting may be distorted. After pickling, the casting is washed in running water.

The casting retrieved from the investment is rough. To remove the roughness, different abrasives are used, viz. silicon carbide, emery, aluminium oxide, garnet, sand, tripoli, rouge, etc. The outer surface of the casting is smoothened with these abrasives using progressively small particle sizes.

Finishing may leave microscopic scratches on the casting. These scratches are eliminated by sequential polishing, producing microcrystalline surface layer, known as *'Bielby layer'*.

Polishing is accomplished with the help of rubber abrasive points, fine particle discs, strips, flexible paper discs, finishing burs and polishing paste. The

Flowchart 16.1: Fabrication of inlays



details of finishing and polishing techniques are described in Chapter 23.

Try in

Remove the temporary restoration and confirm the fit of the casting. A gauze sponge should be placed as a 'throat screen' to catch the casting, if it is accidentally dropped. Do not force the casting on the tooth. If the casting does not seat properly, look for the probable causes. Using the mouth mirror, judge where the proximal contour needs adjustment to allow final seating of the casting. To adjust the proximal contour/contact, several trials on the tooth may be required.

In case of loose contact, soldering can be opted. When the proximal contacts have been adjusted and the casting is satisfactorily seated on the tooth, occlusion is evaluated. Occlusion is verified both in centric occlusion and during lateral excursive movements. Any high points are reduced with suitable abrasive stones.

Cementation

The clinical aspect of cementation is important as many factors play part in successful cementation of the restorations. The cementation procedure is divided into two: Seating and retention.

a. *Seating*

The discrepancies in the prepared tooth and the castings coupled with cementing pressure may lead to rebound of restoration. These discrepancies are to be compensated for proper seating of the restoration. The discrepancies can be minimized by coating the die with die-spacer, and/or relieving the die or etching the casting.

In seating the restoration, the features of luting cement, which plays important role, are:

- Rheology of cement
- Working time
- Film thickness
- Geometry of gap

The failure usually occurs at cement–tooth interface (Fig. 16.38). The flow of cement is the key factor for successfully cementing the restorations. To achieve sufficient cement flow, the seating of restorations should be fast (within a few seconds) while the cement is sufficiently fluid. The flow of zinc phosphate, polycarboxylate and ethoxy-benzoic acid cements is sufficient under moderate pressure. The resin cements are less satisfactory in this respect. Zinc oxide eugenol cement generates the lowest hydraulic pressures during seating, followed by polycarboxylate cement. Zinc phosphate cement exhibited the greatest hydraulic pressure. The seating force should be maintained at least for three minutes without any interruption while the cement is setting (Fig. 16.39A, B).

Working time should be adequate so that the hydraulic action takes place before setting of cement. Mixing on a cool glass slab solves the problem to some extent. The excess cement is to be removed. To achieve easy removal and minimizing the hydraulic pressure, a sort of path is created, known as *venting*. It is confirmed that venting is a satisfactory method of achieving minimal film thickness under cast restorations; it also minimizes the hydraulic pressure.

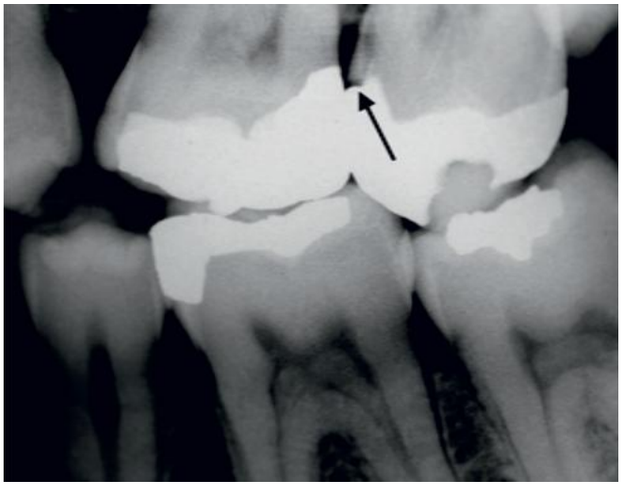


Fig. 16.38: Leakage at cervical margin of inlay (arrow)





Fig. 16.39: (A) Excess cement at the margins of metal-ceramic maxillary anterior crown; **(B)** Excess cement removed from the margins (metal-ceramic posterior crown)

b. *Retention*

It is established that almost all luting cements are fair in compressive strength and poor in tensile strength. The luting agents do not help in retention directly; however, better tensile strength of the cement may aid in retention. The retention of the restoration mainly depends upon the following features:

- Geometrical design of the tooth preparation, which in turn influences the stress distribution.
- Bonding efficiency of the luting cement to both the tooth and the restoration.
- Durability of the cement (resistance to mechanical breakdown and dissolution).

It is recommended that the cement with a high tensile strength be used to cement crowns and long span bridges, since shear stresses in marginal area can exceed the strength of low strength cement.

Viscoelastic behavior might be of more value than brittle characteristics. Dentin exhibits viscoelastic behavior with a significant plastic deformation during fracture, which appears to be greater than that for the zinc phosphate and glass-ionomer cement. And also

the modulus of elasticity of dentin in compression is slightly greater than that of zinc phosphate cements. Bonding between these materials influences the stress transfer between the cement, restoration and the tooth substance.

CASTING DEFECTS

The types of casting defects, commonly encountered by the clinician and the laboratory operator, are summarized in Flowchart 16.2.

Porosity in gold (noble metal) alloy castings is a major defect in casting; hence, described separately. *The possible causes of porosity are as follows.*

a. Porosity Caused by Solidification Defects

i. ***Localized shrinkage porosity:*** When the alloy solidifies from liquid state, it shrinks at least by 1.25%. During solidification of metal in mold, if additional molten metal is not available to compensate for shrinkage, then porosity occurs.

It can occur, if the metal freezes in the sprue before before being pushed in the mold. It generally occurs at the sprue casting junction.

It could also be the result of formation of a hot spot when metal impinges on a mold surface. (*The metal remains molten at this area; while it solidifies everywhere else*.)

This kind of porosity is avoided by:

- Flaring the point of sprue attachment.
- Placing sprue in such a way so that a hot spot formation is avoided.
- Not using an excessively long sprue.
- Using a reservoir.
- Reducing the mold-melt temperature differential (i.e. lowering the casting temperature by about 30°C).

ii. ***Microporosity:*** It occurs due to premature solidification of the metal as a result of solidification shrinkage. The voids produced are irregular in shape. Microporosity is discovered, only if the casting is sectioned. It occurs due to unduly rapid solidification of the metal or casting temperature is too low.

b. Porosity Caused by Trapped Gases

i. Pinhole porosity.
ii. Gas inclusion porosity.

Both produce spherical defects; the defects in case of gas inclusion porosity are larger.

The metal absorbs gases when it is in molten state. Upon solidification, the absorbed gases are expelled and pinhole porosity results.

Larger spherical porosities are caused by gas occluded from a poorly adjusted blowpipe flame or if the reducing zone of the flame is not used.

c. Subsurface Porosity

The exact reason for subsurface porosity has not been established. It may be due to the simultaneous nucleation of solid grains and gas bubbles at the first moment that the metal freezes at the mold walls. This can be diminished by controlling the rate at which the molten metal enters the mold.

d. Back Pressure Porosity (Porosity caused by residual/trapped air)

Porosity produced due to air entrapped on the inner surface of the casting; occurs if for some reason gas is not vented from the mold.

To prevent this, the burnout should be adequate so that carbonaceous residues do not decrease porosity of investment. Mold and casting temperature should not be too low so as to allow metal to solidify rapidly. The water/powder ratio of the investment should be correct. The casting pressure should be adequate.

The rest of the casting defects are tabulated in Table 16.5.

Flowchart 16.2: Casting defects

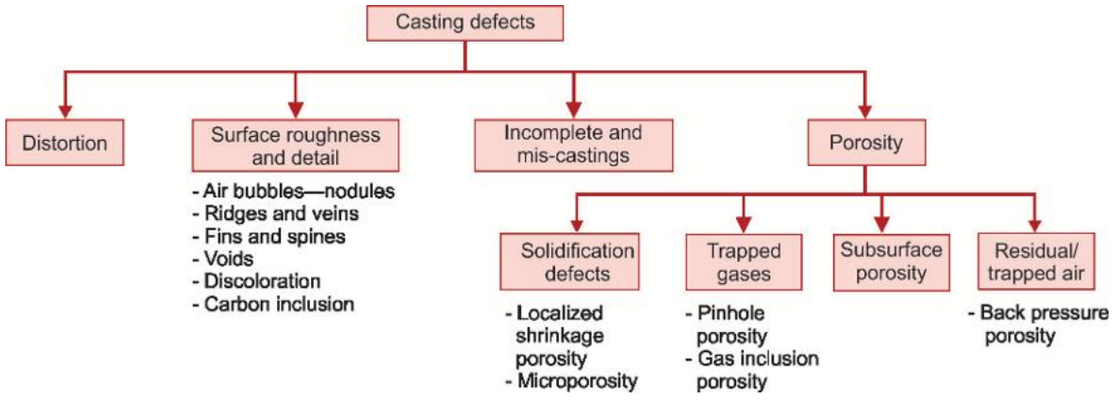


Table 16.5: Casting defects; their causes and solutions

Casting defect	*Causes*	*Solutions*
Incomplete castings	• Inadequate spruing (sprue former too small)	• Use larger size of sprue former. Place sprue former in such a way so that all areas of the mold are fed by molten alloy.
	• Alloy not hot enough	• Have casting temperature at least 150°F (57°C) above fusion temperature of alloy.
	• Mold too cold	• Mold should soak heat for approximately one hour at burnout temperature. The temperature of the oven should be checked regularly with a pyrometer. Mold should be removed from burnout oven and casting completed within one minute.
	• Ingate obstructed	• Ensure that no debris blocks the ingate. The ring should be held with sprue hole down when removing crucible former and sprue former (metal).
	• Insufficient casting force	• Use the adequate amount of force for casting.
	• Insufficient gold	• Cast enough gold to allow for a good button in the crucible of the ring.
Rounded margins	• Incomplete burnout of wax pattern	• Carbon residues in the mold can produce shiny rounded margins. Ensure adequate burnout time and temperature facilitating carbon to get converted to carbon monoxide or carbon dioxide.
	• Insufficient heating of alloy before casting	• Heat alloy to 150°F (57°C) above fusion temperature.
	• Improper diameter/length of sprue restricts flow of alloy into the mold. The metal freezes before margins are complete.	• Average casting should have 10 gauge sprue and approximately 6.0 mm long.
Pits in casting	• Debris in mold	• Join the sprue former, crucible former and pattern with continuous smooth surface with no jagged areas of investment, which can break off and enter mold.
	• Dirty wax	• Use clean, new wax for patterns and sprues.
	• Loose debris in crucible	• Use clean crucible for each casting
	• Mold temperature too hot	• At very high temperature, the investment can break. Avoid overheating.
Distortion	• Distortion of wax pattern	• Proper handling and manipulation of the wax pattern.
	• Due to uneven movement of the pattern when the investment is setting. The gingival margins are forced apart by the mold expansion, whereas the solid occlusal bar of wax resists expansion during the early stage of setting.	• The pattern should be gently and uniformly invested.
Surface roughness, irregularities	• Air bubbles on the pattern	• Use vacuum investing technique. • Vibrate before and after mixing. • Use a wetting agent to reduce surface tension of wax pattern. • Air dry the wetting agent as excess water will dilute investment, causing irregularities.
	• Water films causing ridges and veins on the surface	• Avoiding movement, vibration of pattern after investment. Painting investment properly on the pattern to ensure intimate contact. • Using a wetting agent.

(*Contd.*)

Table 16.5: Casting defects, their causes and solutions (*Contd.*)

Casting defect	*Causes*	*Solutions*
	• Too rapid heating	• The mold should be heated gradually, 60 minutes taken to heat in fins or spines the ring from room temperature to 700°C. • Rapid heating will cause flaking of investment when steam forms. Also a surge of steam may carry salts used as modifiers into the mold.
	• Underheating, causing incomplete elimination of wax	• Heat the ring for adequate period of time so that the carbonaceous residue is removed.
	• Inappropriate water: Powder ratio	• Both excess W/P and low W/P ratio causes rough castings. Thus W:P should be accurate. Low W:P ratio causes roughness because it cannot be painted on properly and during vacuum investing air might not be removed completely.
	• Prolonged heating	• Gypsum-bonded investment should never be heated above 700°C because sulfur compounds are formed and investment starts disintegrating.
	• Temperature of alloy too high	• The color of the molten alloy just prior to casting should be no lighter than light orange. Very hot alloy will attack surface of investment causing roughness.
	• Casting pressure too high	• Pressure required for casting should be adjusted.
	• Foreign bodies	• Avoid carrying pieces of investment loosened during removal of sprue former and crucible former. • A piece of carbon from the flux could be carried into the mold to cause a bright appearing concavity.
	• Impact of molten alloy	• The direction of sprue former should be adjusted so that it does not strike a weak part of the mold surface at an angle of 90°.
	• Pattern position	• Several patterns when placed in the same ring should not be placed too close, because expansion of wax may lead to breakdown of investment, if proper thickness of investment is not present.
Discoloration	• Sulfur contamination of casting, causing black castings	• This could be due to (a) heating the investment above 700°C causing its breakdown and formation of sulfur compounds. It can be avoided by avoiding overheating of investment. (b) sulfur content of the torch flame. The problem can be solved by changing the source of heat.
	• Contamination with iron/copper during pickling	• Avoiding the use of steel tongs to hold casting during pickling. If they must be used, then the tips of the tongs should be covered with rubber.
	• Contamination with mercury	• Castings should never be placed with amalgam dies or kept on a table where amalgam scrap is present.

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